Facet: Prioritized Population-Based Reinforcement Learning for Human-Compatible Multiplayer Agents

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Abstract

AI agents are increasingly deployed in *multiplayer workspaces*: shared environments in which several agents and several humans act concurrently toward a common objective, as in Jewel Labs' agentic infrastructure for the diamond and jewelry trade. Reinforcement-learning agents trained by self-play excel when paired with copies of themselves but are known to fail when paired with partners—human or artificial—whose conventions differ from their own. We present Facet (Fictitious Agent Co-training for Emergent Teamwork), a lightweight population-based training method that (i) seeds the partner population with behavior-cloned human models fit to real human gameplay, (ii) adds frozen self-play checkpoints spanning multiple skill stages, and (iii) prioritizes sampling toward the partners with which the learner currently coordinates *worst*. On the Overcooked-AI benchmark, using the 2019 human–human dataset (103,141 timesteps across five layouts), we train and evaluate 75 models: behavior cloning (BC), self-play PPO (SP), PPO trained with a fixed human model (PPO_{BC}), and Facet (three seeds each per layout), plus 15 held-out human-proxy models—and evaluate every agent along two axes of the workspace problem: coordination with *held-out human proxies* and with an *existing fleet* of independently trained agents (in-distribution for Facet by construction, unseen for the baselines). The classic failure reproduces sharply at laptop scale: self-play loses 53–96% of its self-paired score when moved to human proxies. No single method dominates both axes—PPO_{BC} is the strongest human-proxy method (best of the four methods on 3/5 layouts) but sacrifices fleet compatibility, while self-play shows the reverse profile. Across the ten seed-averaged (layout × partner-type) evaluation cells, Facet is the only method without a catastrophic cell: its worst cell scores 12.7 versus 3.7–5.8 for all baselines, and it attains the best combined human+fleet score under equal axis weighting (110.3 vs. 102.0 for PPO_{BC} and 95.0 for SP; Section 6 discusses both qualifications). All code, all 75 trained models, and an interactive browser demo are released.[1]

1 Introduction

Enterprise software is shifting from single-user tools to *multiplayer workspaces*: shared, real-time environments in which human teams and autonomous AI agents act concurrently—humans monitoring, redirecting, and approving while agents quote, plan, and verify. Jewel Labs' production

*Jewel Labs Research — https://jewellabs.org.

[1] https://github.com/aarushkandukoori/facet

platform for the diamond trade is one such workspace: quoting, cut-planning, and compliance agents operate alongside human traders in a shared operational surface. In such settings the value of an agent is not its solo competence but its *team* competence: the joint performance of the human–agent system [13]. This echoes a 25-year-old lesson from mixed-initiative interaction design—automation is useful exactly insofar as it couples smoothly with human activity [5].

Reinforcement learning offers a direct way to train agents for interactive multi-step tasks, and *self-play*—training an agent with copies of itself—is its workhorse. But a well-replicated finding in cooperative multi-agent RL is that self-play agents develop private conventions that shatter when the partner deviates from them. Carroll et al. [2] demonstrated this on Overcooked, a two-player common-payoff coordination game (which we re-skin as a diamond-cutting atelier in Section 3, matching our deployment domain): agents trained by self-play or population-based training scored highly with themselves yet often dropped below a simple imitation baseline when paired with human proxies. Hu et al. [6] formalized the underlying issue as *zero-shot coordination* (ZSC): converging to arbitrary symmetric conventions is optimal in self-play and catastrophic with novel partners.

Two families of remedies exist. *Human-data* methods embed a learned human model in training: Carroll et al. [2]'s PPO_{BC} trains a best response to a behavior-cloned human. *Population* methods avoid human data by training a best response to a diverse population: TrajeDi originated the diversity-regularized population plus common-best-response recipe [12]; Fictitious Co-Play (FCP) [17] showed that independent self-play runs *and their past checkpoints* suffice as partners; later work refines the population via entropy bonuses [26], hidden-utility biases [24], or generative partner models [9].

These two families are usually studied in opposition—with human data or without. In a deployed multiplayer workspace, however, both resources exist simultaneously: logs of real human behavior *and* cheap self-play checkpoints. FACET combines them. The partner population contains behavior-cloned human models (convention anchors from real data) and frozen self-play checkpoints from independent runs at four skill stages (cheap convention and skill diversity, as in FCP). Training then applies a *prioritized sampling* rule inspired by MEP's second stage [26]: partners are drawn with probability increasing in the learner's current *failure* to coordinate with them, estimated by an exponential moving average of episode returns. The learner therefore spends its samples where its coordination is weakest—typically the human models and the low-skill checkpoints—while never fully abandoning strong partners.

We make three contributions:

1. **Method.** FACET, a simple best-response-to-population scheme unifying human behavior models and self-play checkpoints under worst-partner-prioritized sampling (Section 4).

2. **Controlled two-axis study at laptop scale.** A complete, reproducible comparison of BC, SP, PPO_{BC}, and FACET—60 RL/imitation policies plus 15 held-out human-proxy models—trained on a single Apple M4 laptop in under six hours total, on all five layouts of the classic Overcooked human–human dataset [2], and evaluated against both held-out human proxies and an existing agent fleet (Section 5). All numbers in this paper are measured from these runs; no results are imported from prior work.

3. **Open release.** Training code, all 75 model checkpoints, evaluation data, and an interactive in-browser replay demo of the trained agents.[2]

Our headline results (Tables 2 and 3): the self-play failure reproduces dramatically—SP loses between 53% (Cramped Room) and 96% (Forced Coordination) of its self-paired score when paired

[2] https://github.com/aarushkandukoori/facet

with held-out human proxies. PPO_{BC} is the strongest single-axis human-compatibility method (its mean exceeds FACET's on four of five layouts on that axis), but it pays for that specialization on the machine-partner axis, where FACET's mean exceeds PPO_{BC}'s on all five layouts—an axis that is in-distribution for FACET by construction, since its training pool contains the fleet runs' checkpoints (Section 5). No method wins everywhere; what the multiplayer-workspace setting arguably demands is the *absence of catastrophic cells*, and on seed-averaged cells FACET is alone: its worst of the ten (layout × partner-type) cell means is 12.7, versus 3.7 for SP, 5.3 for PPO_{BC}, and 5.8 for BC, and its combined human+fleet average (110.3, equal axis weighting) is the highest of the four methods. Individual seeds still fail—we report per-seed minima and self-pairing deadlocks for *all* methods, including FACET, in Section 6.

2 Related Work

Overcooked and human-compatible RL. Carroll et al. [2] introduced the two-player Overcooked benchmark, collected the human–human dataset we use, and showed that self-play PPO and population-based training degrade sharply when paired with human proxies while PPO_{BC} (best response to a behavior-cloned human) transfers better; their user study (N=60) confirmed the effect with real humans. The benchmark has since become the standard testbed for human–AI coordination. Follow-up work has questioned aspects of its evaluation: ZSC-Eval [19] proposes best-response-diversity-based partner selection, and OvercookedV2 [3] argues the original layouts under-test ZSC because broad state coverage suffices; we therefore report per-layout results and self-pair/cross-pair gaps rather than a single aggregate.

Population-based zero-shot coordination. Other-Play [6] framed ZSC and showed self-play converges to arbitrary conventions. TrajeDi [12] introduced the diversity-regularized population plus common-best-response recipe; FCP [17] showed a population of self-play runs *and their checkpoints* suffices to coordinate with humans without human data; MEP [26] added a population-entropy bonus and *prioritized* partner sampling; HSP [24] argued populations should span human *biases*, not just skill; E3T [21] questioned the cost of populations altogether; GAMMA [9] replaced the discrete population with a generative partner model, optionally biased by a small amount of human data. PECAN [11] and COLE [8] further refine population construction and pairing, and unsupervised environment design offers diversity over *environments* rather than partners [22]. FACET borrows FCP's checkpoint zoo, MEP's prioritization, and PPO_{BC}'s human anchor, in deliberately minimal form: no diversity bonus, no generative model, one best-response run.

MARL foundations. Our learners are PPO [16] with GAE [15]; parameter-shared PPO is a strong cooperative MARL baseline [23]. The human models are behavior cloning [1, 14] over the benchmark's hand-designed 96-dimensional featurization.

Human–AI teaming and LLM-agent workspaces. The National Academies consensus report argues AI should be evaluated as a *teammate*, by combined team effectiveness rather than standalone capability [13]. Recent LLM multi-agent frameworks—AutoGen [20], CAMEL [7], MetaGPT [4]—orchestrate agent–agent collaboration but largely treat the human as an input source rather than a partner to be modeled; LLM agents are also entering Overcooked itself, via proactive cooperators [25], hierarchical real-time agents [10], and process-oriented benchmarks [18], which find that current LLMs "exhibit significant shortcomings in active collaboration and continuous adaptation." FACET addresses the complementary control layer: fast reactive policies that must mesh with human behavior at every timestep, the regime where per-step LLM inference remains impractical.

3 Preliminaries

Setting. A two-player common-payoff Markov game $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, r, H)$: at state s_t both players select actions $(a_t^0, a_t^1) \in \mathcal{A}^2$, the state transitions deterministically, and both receive the same reward r_t. A policy pair (π^0, π^1) earns expected return $J(\pi^0, \pi^1) = \mathbb{E}\big[\sum_{t=0}^{H-1} r_t\big]$. Self-play maximizes $J(\pi_\theta, \pi_\theta)$; the quantity that matters for deployment is $J(\pi_\theta, \pi_H)$ for the actual partner distribution π_H, which at training time is unknown.

Overcooked as a diamond atelier. Our testbed is the Overcooked benchmark [2], which we present throughout this paper—and render in the released demo—under a *diamond-atelier skin* matched to Jewel Labs' domain: two workers in a gridworld workshop must produce and deliver finished diamonds. A worker fetches *rough stones* (onions, in the benchmark's original kitchen skin) and loads three of them into a *cutting wheel* (pot); after 20 timesteps of processing, the finished *diamond* (soup) is collected onto a *presentation tray* (dish) and handed over at the *client counter* (serving window) for a shared reward of 20 per delivery. Only the vocabulary and rendering are ours; the dynamics, layouts, and human data are the unmodified benchmark, so every number remains comparable to the Overcooked literature. Episodes last $H = 400$ timesteps. Movement is simultaneous; workers collide, block corridors, and must pass stones over counters on some layouts. We use all five classic layouts (Figure 1): *Cramped Room* (tight quarters, collision avoidance), *Asymmetric Advantages* (asymmetric roles), *Coordination Ring* (traffic direction around a loop), *Forced Coordination* (neither worker can complete a diamond alone— stones and trays must be passed across a divider), and *Counter Circuit* (long circuit; passing stones over the central counter is the expert strategy). Following the benchmark's human-data configuration, we use the 2019 ("old") dynamics in which the wheel starts automatically when the third stone is loaded. Observations are the benchmark's 96-dimensional egocentric featurization $\phi(s, i)$; the action space has six discrete actions (four moves, stay, interact).

Human data. We use the cleaned 2019 human–human trials shipped with the benchmark: 103,141 joint timesteps across the five layouts, split by whole games into a training set (63 games, 75,744 timesteps) and a held-out test set (23 games, 27,397 timesteps); each joint timestep yields two (observation, action) examples, one per seat (Table 1).

4 Method

All four methods share one policy class: a two-hidden-layer MLP (128 units, tanh) mapping $\phi(s, i)$ to action logits and a value estimate (Section 5). We compare:

BC. Behavior cloning: $\pi_{\mathrm{BC}} = \arg\max_\pi \sum_{(s,i,a)\in\mathcal{D}_{\mathrm{train}}} \log \pi(a \mid \phi(s, i))$, a 64-unit MLP classifier trained with early stopping on a validation split. Three seeds per layout. BC models act *stochastically* (sampling from the softmax), which preserves human-like variability [2].

SP. Parameter-shared self-play PPO: one network controls both seats; both seats' transitions enter the buffer. This is the standard strong cooperative baseline [23].

Table 1: The 2019 Overcooked human–human dataset [2] as used here: whole-game train/test split; each joint timestep contributes one sample per seat. BC human models are fit on the train split; the evaluation proxies H_{proxy} are fit on the test split.

	Train split		Test split (H_{proxy})	
Layout	games	timesteps	games	timesteps
Cramped Room	14	16,788	5	6,013
Asymm. Advantages	14	16,851	5	5,942
Coordination Ring	13	15,639	5	5,950
Forced Coordination	9	10,831	3	3,544
Counter Circuit	13	15,635	5	5,948
Total	63	75,744	23	27,397

Algorithm 1 FACET training

input BC models $\mathcal{P}_{\text{BC}}$, SP checkpoints $\mathcal{P}_{\text{ckpt}}$, temperature τ, floor ε
$\mathcal{P} \leftarrow \mathcal{P}_{\text{BC}} \cup \mathcal{P}_{\text{ckpt}}$; initialize learner π_θ; $\tilde{R}_i \leftarrow \varnothing$
for each parallel environment e **do**
  assign learner to seat e mod 2; sample partner $i_e \sim p$ ▷ Eq. (1)
end for
while steps < budget **do**
  collect rollouts; learner acts with π_θ, seat-partner acts with frozen π_{i_e}
  for each finished episode in environment e with sparse return R **do**
    $\tilde{R}_{i_e} \leftarrow (1-\beta)\tilde{R}_{i_e} + \beta R$; re-sample $i_e \sim p$
  end for
  update θ with PPO on the learner's transitions only
end while

PPO$_{\text{BC}}$. PPO best response to a fixed sampled-BC partner (one BC seed per learner seed), the human-aware baseline of Carroll et al. [2]. The learner occupies each seat in half the parallel environments.

Facet. PPO best response (Algorithm 1) to a *prioritized population* $\mathcal{P} = \mathcal{P}_{\text{BC}} \cup \mathcal{P}_{\text{ckpt}}$: the three BC human models, plus every checkpoint of the three independent SP runs saved at 8%, 20%, 40%, and 100% of training (12 checkpoints), giving $|\mathcal{P}| = 15$ frozen partners spanning human conventions and machine skill stages. Each parallel environment holds a partner sampled from

$$p_i = \frac{\varepsilon}{|\mathcal{P}|} + (1-\varepsilon)\frac{\exp\big(-\bar{R}_i/\tau\big)}{\sum_j \exp\big(-\bar{R}_j/\tau\big)}, \qquad \bar{R}_i = \frac{\tilde{R}_i - \min_j \tilde{R}_j}{\max(\max_j \tilde{R}_j - \min_j \tilde{R}_j,\, 1)}, \tag{1}$$

where $\tilde{R}_i$ is an exponential moving average (decay 0.05) of the sparse episode return achieved with partner i, re-sampled at every episode boundary; unseen partners take $\bar{R}_i = 0$ (maximum priority). We use $\tau = 0.35$ and $\varepsilon = 0.25$ throughout. The rule concentrates experience on the partners the learner currently fails with—empirically the BC models and early checkpoints—while the ε-floor prevents catastrophic forgetting of strong partners (Figure 2). FACET costs one PPO run plus reuse of artifacts that self-play training already produced.

Reward. All RL methods optimize the shared sparse reward (+20 per delivered diamond) plus the benchmark's per-player shaped rewards (stone loaded into the wheel +3, tray pickup +3, finished-diamond pickup +5) scaled by a coefficient annealed linearly from 1 to 0 over the first 60% of training, so final policies are shaped only by task reward. Rewards are scaled by 0.1 for optimization; all reported scores are raw sparse returns.

5 Experimental Setup

Implementation. PPO [16] with GAE ($\gamma = 0.99$, $\lambda = 0.95$) [15], clip 0.2, 8 epochs of minibatch 2048 per update, Adam at 3×10^{-4}, entropy coefficient annealed $0.05 \to 0.005$ over 80% of training, gradient-norm clip 0.5. Rollouts come from 15 environments (3 worker processes $\times$ 5 serial environments) of horizon 400. Policies are $96 \to 128 \to 128 \to 6$ MLPs with a separate identical value head ($\sim$59k parameters total); BC models are $96 \to 64 \to 64 \to 6$. Training budget is 5M environment steps per run on Cramped Room, Asymmetric Advantages, and Coordination Ring, and 7M on the harder Forced Coordination and Counter Circuit. Every number in this paper was produced on one Apple M4 MacBook (10 cores, no GPU); measured throughput is $\sim$4.0–4.9k environment steps/second for SP runs, $\sim$4.1–5.9k for PPO_{BC}, and $\sim$2.9–4.2k for FACET (the population adds frozen-partner inference), and the full 45-run RL suite plus evaluation completes in under six hours. Complete hyperparameters in Appendix A.

Evaluation protocol. Following Carroll et al. [2], coordination with humans is estimated by pairing each agent with H_{proxy}: sampled-BC models trained on the *held-out test games*, which no agent saw in any form during training. BC (train-split), SP, PPO_{BC}, and FACET agents (3 seeds each) are each paired with all 3 H_{proxy} seeds for 10 episodes per pairing—5 per seat assignment—giving 90 episodes per method–layout cell. We also report each agent paired with itself (30 episodes per cell). We report the mean over episodes, with standard errors computed over the 3 training seeds (per-seed means as units). As a human-team reference we also report the mean score of the actual human–human test games, normalized to the 400-step horizon (the human games ran $\approx$1,190 steps).

Fleet cross-play. Multiplayer workspaces contain other *agents*, not only humans, so we additionally pair every policy with the three final SP agents, treated as an existing deployed fleet (10 episodes per pairing; same-seed SP pairings excluded, as those are the self-pair cells). One disclosure matters for interpretation: FACET's partner pool contains checkpoints of exactly these SP runs—including the 100% checkpoints, i.e. the fleet agents themselves—so for FACET this axis measures *retained* in-distribution fleet compatibility rather than zero-shot transfer—which is the realistic deployment question (a new agent must join a fleet that is known at training time), while for SP, PPO_{BC}, and BC the fleet partners are unseen. A second disclosure: on Counter Circuit one of the three fleet agents is the degenerate SP seed that never learned to score (Figure 1), which depresses that row of Table 3 for every method. Per-seed scores for the human-proxy axis are in Appendix B. Cross-seed SP+SP pairing is the classic zero-shot convention-clash test [6].

6 Results

Self-play collapses when the partner changes. Table 2 shows the classic result cleanly at our scale. SP posts the highest self-paired scores everywhere (up to 406.0 on Asymmetric Advantages) yet drops 53% (Cramped Room), 79% (Coordination Ring), 85% (Asymmetric Advantages), 86% (Counter Circuit), and 96% (Forced Coordination) when moved to held-out human proxies. On Forced Coordination—where neither worker can complete a diamond alone—SP with a human

Table 2: Mean episode score (delivered diamonds × 20, horizon 400) when paired with held-out human proxies (H_{proxy}), and when paired with self. Mean ± s.e. over 3 training seeds (90 evaluation episodes per H_{proxy} cell, 30 per self cell). *HH* = mean score of the real human–human test games, rate-normalized from ≈1,190-step games to the 400-step horizon (a reference, not an upper bound). Bold = best with H_{proxy} per layout, plus any method whose ±1 s.e. interval overlaps the best's.

	BC	SP	PPO_{BC}	FACET	HH
Cramped Room	58.9 ± 2.5	**103.3 ± 5.1**	**102.7 ± 2.7**	**90.2 ± 18.6**	25
with self	76.7 ± 4.7	220.0 ± 1.2	0.0 ± 0.0	70.7 ± 70.7	
Asymm. Advantages	68.2 ± 4.7	60.0 ± 8.3	**136.9 ± 16.8**	84.4 ± 6.1	40
with self	74.0 ± 8.1	406.0 ± 8.1	284.7 ± 15.8	279.3 ± 42.7	
Coordination Ring	21.8 ± 1.4	**37.6 ± 1.6**	**57.6 ± 31.0**	**27.3 ± 3.1**	21
with self	29.3 ± 0.7	181.3 ± 5.3	59.3 ± 58.3	0.0 ± 0.0	
Forced Coordination	8.2 ± 1.2	5.8 ± 4.5	**39.1 ± 2.2**	**35.3 ± 1.7**	27
with self	16.7 ± 4.4	160.0 ± 0.0	107.3 ± 5.3	113.3 ± 3.7	
Counter Circuit	**17.6 ± 1.0**	**12.0 ± 6.4**	5.3 ± 1.4	**12.7 ± 5.4**	13
with self	22.0 ± 4.2	84.7 ± 42.4	0.0 ± 0.0	4.0 ± 4.0	

Table 3: Fleet cross-play: mean episode score when paired with the three final SP agents (10 episodes per pairing; mean ± s.e. over 3 agent seeds; same-seed SP pairings excluded). FACET's pool contained these runs' checkpoints (retained compatibility); for the other methods the fleet is unseen. Bold = best per layout, plus any method whose ±1 s.e. interval overlaps the best's.

	BC	SP	PPO_{BC}	FACET
Cramped Room	114.7 ± 1.7	**206.7 ± 4.3**	136.4 ± 6.8	**203.6 ± 6.3**
Asymm. Advantages	66.7 ± 6.0	**396.7 ± 2.3**	322.4 ± 8.8	338.9 ± 19.5
Coordination Ring	51.6 ± 2.5	84.7 ± 23.2	123.1 ± 16.4	**141.3 ± 1.3**
Forced Coordination	5.8 ± 0.6	3.7 ± 1.8	82.2 ± 10.6	**132.2 ± 3.7**
Counter Circuit	**19.6 ± 0.6**	**39.3 ± 19.7**	**13.8 ± 9.4**	**36.7 ± 8.7**

proxy (5.8) is *worse than plain BC* (8.2), reproducing Carroll et al. [2]'s central observation that self-paired performance is not predictive of human-paired performance.

Human anchoring is the single strongest ingredient. PPO_{BC} is the best human-proxy method on three of five layouts (Asymmetric Advantages 136.9, Coordination Ring 57.6, Forced Coordination 39.1) and ties SP on Cramped Room (102.7 vs. 103.3). At this compute scale, concentrating all best-response capacity on a single human model transfers to unseen human proxies better than spreading it across a 15-partner population: PPO_{BC}'s mean exceeds FACET's on four of five layouts on this axis (on Cramped Room the 102.7-vs-90.2 gap is within FACET's seed s.e. of ±18.6), though FACET clearly beats SP where SP collapses hardest (84.4 vs. 60.0 on Asymmetric Advantages; 35.3 vs. 5.8 on Forced Coordination) and trails PPO_{BC} on Forced Coordination by less than the sum of their s.e.s. Counter Circuit is sobering for every RL method: BC itself (17.6) is the top proxy-paired policy, echoing the original paper's finding on this layout.

Specialists pay on the other axis. Table 3 completes the picture. PPO_{BC}'s human specialization costs it with machine partners: 136.4 vs. FACET's 203.6 on Cramped Room, 82.2 vs. 132.2 on Forced Coordination, 13.8 vs. 36.7 on Counter Circuit. The specialization runs deep enough that PPO_{BC} is not even compatible with *itself*: two copies of the same PPO_{BC} agent score 0.0 on Cramped Room and Counter Circuit (Table 2, *with self* rows) while the same agents score 102.7

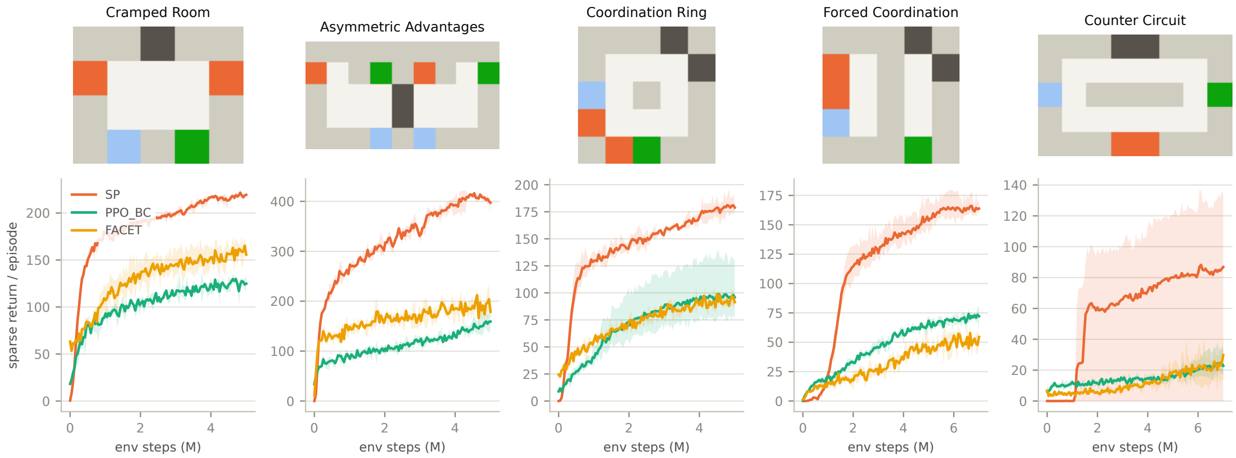

Figure 1: Top: the five layouts (orange = rough-stone dispensers, blue = tray dispensers, dark = cutting wheels, green = client counters). **Bottom:** training curves (sparse return per episode; mean over 3 seeds, min–max band). Curves are measured against each method's *training* partners and are not comparable across methods—SP plays with a copy of itself, $\mathrm{PPO}_{\mathrm{BC}}$ with a stochastic human model, FACET with its mixed pool. One SP seed on Counter Circuit never scored (flat 0 for 7M steps), a known failure mode of self-play on this layout.

and 5.3 with human proxies—both copies occupy the role the human model used to fill, a concrete instance of the cooperative incompatibility studied by Li et al. [8]. We verified this is genuine deadlock rather than inactivity: the agents move and interact throughout the episode without ever completing a delivery chain. Importantly, *FACET exhibits the same failure*: its self-pairing scores 0.0 on Coordination Ring (all three seeds), and its Cramped Room (70.7) and Counter Circuit (4.0) self-pair means hide bimodal per-seed outcomes (212.0/0.0/0.0 and 0.0/12.0/0.0 respectively)—best-response training against any external partner distribution, human or mixed, does not confer self-compatibility. Most self-pair cells for both best-response methods are all-or-nothing per seed, which the s.e.s in the *with self* rows make visible. Cross-seed SP+SP pairing is strong where independent runs converge to similar conventions (396.7 on Asymmetric Advantages) and catastrophic where they do not (3.7 on Forced Coordination—a pure convention clash, since each SP agent scores ≈160 with its own copy). FACET is best or within one s.e. of best on four of five layouts, with visibly lower seed variance in several cells (141.3 ± 1.3 on Coordination Ring vs. 84.7 ± 23.2 for SP).

No catastrophic seed-averaged cells. Averaging the two axes per layout and then across layouts gives a combined workspace-compatibility score of 110.3 for FACET, 102.0 for $\mathrm{PPO}_{\mathrm{BC}}$, 95.0 for SP, and 43.3 for BC (equal axis weighting; see Section 7). More telling than the average is the minimum: across all ten seed-averaged (layout × partner-type) cells, FACET's worst cell mean is 12.7, whereas SP bottoms out at 3.7, $\mathrm{PPO}_{\mathrm{BC}}$ at 5.3, and BC at 5.8 (Figure 3 visualizes the self-vs-proxy gaps). This is a property of 3-seed means, not of every agent: FACET's worst single seed scores 3.3 (Counter Circuit, seed 0, with H_{proxy}; Appendix B), so individual FACET agents can still fail, and three seeds is a thin basis for a robustness claim. In a deployed workspace, the worst cell is the one users remember—and seed selection against a validation partner would be part of any real deployment.

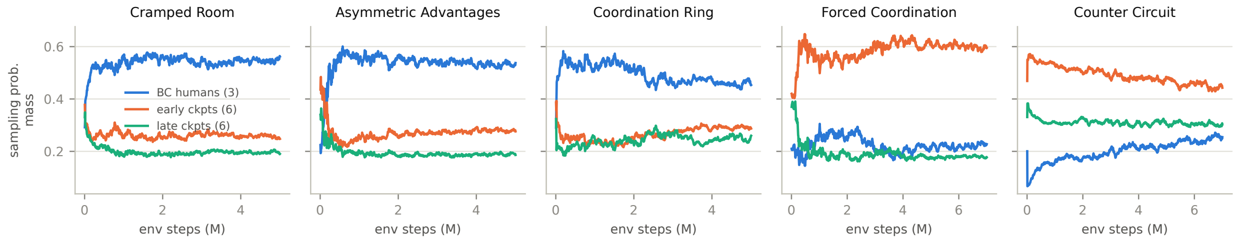

Figure 2: Prioritized sampling mass over FACET training (seed 0), grouped by partner type: BC human models (3 partners), early SP checkpoints (8%/20%, 6 partners), late SP checkpoints (40%/100%, 6 partners). The sampler concentrates on human models where they are the hardest partners (left three layouts) and on low-skill checkpoints on the two hard layouts.

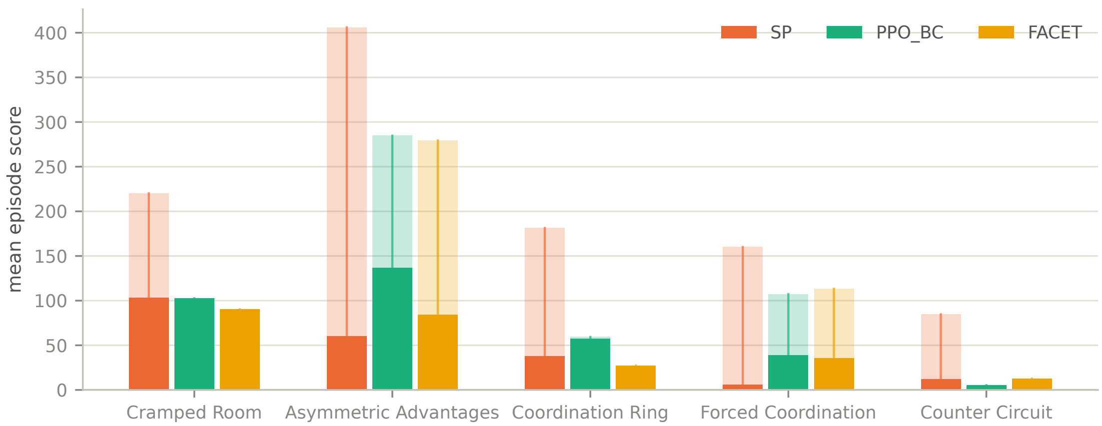

Figure 3: Self-paired score (pale) vs. score with held-out human proxies (solid) for the three RL methods. The pale-to-solid drop is the generalization gap that self-play maximizes and population/human-anchored methods shrink.

The sampler finds the hard partners. Figure 2 tracks the prioritized sampler's probability mass during FACET training. On the three easier layouts, mass concentrates on the three BC human models (49–54% of sampling, vs. 20% under uniform sampling)—the human models are the hardest partners there. On Forced Coordination and Counter Circuit the mass shifts to the early low-skill checkpoints instead: with an incompetent partner on these layouts, no diamond can be delivered at all, making those the lowest-return partners. The prioritization thus adapts the curriculum per layout without any manual tuning, at the cost of spending fewer samples on the human models exactly on the layouts where human compatibility is hardest—a plausible explanation for FACET's remaining gap to PPO_{BC} on the human axis, and a concrete lever for future work.

7 Discussion and Limitations

What the results support. Three conclusions are directly supported. First, the classic finding reproduces at laptop scale: self-paired scores are never predictive of human-paired scores, with drops of 53–96%. Second, at this compute scale human-data anchoring is the strongest single ingredient for the human axis—PPO_{BC} beats the population method on three of five layouts, a useful negative result given that population methods dominate recent literature at much larger scale [17, 26]; our population is small (15 partners from 3 runs) and a best response to it necessarily dilutes capacity relative to a single-partner specialist. Third, specialization has a measurable price on the other axis, and only the population method avoids catastrophic cells across both axes. We would summarize the practical recipe for multiplayer workspaces as: *if you will only ever face humans, best-respond to a human model; if you must coexist with a fleet as well, FACET-style pools buy worst-case robustness for one extra PPO run.*

What the results do not support. Our results do not show that prioritized pools beat human-anchored best response on the human axis at this scale—they do not, on three of five layouts. We report this directly rather than aggregating it away; the combined score (110.3 vs. 102.0) favors FACET only because the fleet axis is weighted equally, which is a modeling choice about deployment, not a finding.

Limitations. First, our human-side evaluation uses BC proxies, not live humans; proxy evaluation is the field's standard first filter but is an imperfect predictor of performance with real humans [17, 19], and a user study remains future work. Second, all agents use the benchmark's hand-designed featurization and old dynamics to remain commensurate with the 2019 human data; results may differ under pixel observations or the newer dynamics. Third, our compute scale is deliberately small—one laptop—so absolute scores trail large-scale reproductions; the comparisons are internally controlled (identical architecture, budget, and evaluation for every method), but ceilings may shift with more compute. Fourth, Overcooked itself under-tests some coordination phenomena [3]; extending FACET to richer workspace simulators—including the tool-use and approval-gate structure of real enterprise workspaces—is the natural next step for Jewel Labs' production setting. Fifth, our fleet axis is in-distribution for FACET by construction—its pool contains the fleet agents' checkpoints, including the final ones—so Table 3 measures retained compatibility with a known fleet, not zero-shot transfer to unknown agents; a held-out-fleet evaluation (e.g., SP runs excluded from the pool) would isolate the latter and we did not run one. Sixth, the released browser demo replays each pairing's best episode from its best seed—it is an illustration of learned behavior, not a random sample of the means in Table 2. Finally, FACET inherits FCP's main cost: it consumes the checkpoints of multiple self-play runs (plus a $\sim$15–25% per-step overhead for frozen-partner inference), though the checkpoints are artifacts most practitioners already produce.

8 Conclusion

FACET is a deliberately minimal recipe—human-anchored population, checkpoint skill diversity, worst-partner prioritization—trained and evaluated end-to-end on a laptop across all five classic Overcooked layouts. The study's clearest lessons are that self-play's 53–96% partner-transfer collapse survives every scale reduction; that a best response to one behavior-cloned human remains the strongest pure human-compatibility baseline at small scale; and that a prioritized pool of human models plus self-play checkpoints is what removes the catastrophic seed-averaged cells when agents must work with humans *and* a fleet that is known at training time—the defining condition of a multiplayer workspace. For organizations building such workspaces, the ingredients are already lying around: behavior logs, self-play checkpoints, and a ten-line prioritized sampler.

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A Hyperparameters

Table 4: Complete hyperparameters. All methods share the PPO block; BC uses the imitation block. No per-layout tuning was performed.

PPO (SP, PPO$_{\mathbf{BC}}$, Facet)		**Behavior cloning**	
policy net	MLP 96−128−128−6, tanh	net	MLP 96−64−64−6, ReLU
value net	MLP 96−128−128−1, tanh	optimizer	Adam, lr 10^{-3}
optimizer	Adam, lr 3×10^{-4}	batch size	512
discount γ / GAE λ	0.99 / 0.95	max epochs	60
PPO clip	0.2	early stop	patience 8 (val. loss)
epochs × minibatch	8×2048	val. fraction	0.1
entropy coef.	$0.05\to0.005$ (80%)	weight decay	10^{-6}
value coef. / grad clip	0.5 / 0.5		
reward scale	0.1	**Facet population**	
shaped-reward anneal	$1\to0$ over 60%	pool	3 BC + 12 SP ckpts
parallel envs	15 (3×5), horizon 400	ckpt stages	8/20/40/100%
steps (easy / hard layouts)	5M / 7M	τ, ε, EMA β	0.35, 0.25, 0.05

B Per-seed results

C Reproducibility

All experiments are reproducible from the released repository: `python -m facet.data` and `python -m facet.bc` build the datasets and BC models (∼25 minutes); `scripts/run_batch.sh` executes the PPO stages via `scripts/queue_sp.txt` then `scripts/queue_br.txt` (∼5 hours on a 10-core M4; the queue files carry the paper's step/worker settings—`train_ppo.py`'s bare defaults are smaller); `scripts/evaluate.py` and `scripts/eval_fleet.py` produce the raw episode scores per layout; and `scripts/analyze.py` (`results`, `fleet`, `figures`) regenerates every table and figure from them. Random seeds are fixed (0, 1, 2) at every stage.

Table 5: Per-seed mean scores with H_{proxy} (30 episodes per seed: 3 proxy seeds $\times$ 10 episodes).

Layout	Method	seed 0	seed 1	seed 2
Cramped Room	BC	60.7	62.0	54.0
Cramped Room	SP	106.7	93.3	110.0
Cramped Room	PPO_{BC}	108.0	99.3	100.7
Cramped Room	FACET	127.3	71.3	72.0
Asymm. Advantages	BC	75.3	70.0	59.3
Asymm. Advantages	SP	73.3	62.0	44.7
Asymm. Advantages	PPO_{BC}	107.3	138.0	165.3
Asymm. Advantages	FACET	96.7	77.3	79.3
Coordination Ring	BC	24.0	19.3	22.0
Coordination Ring	SP	34.7	40.0	38.0
Coordination Ring	PPO_{BC}	30.7	22.7	119.3
Coordination Ring	FACET	25.3	23.3	33.3
Forced Coordination	BC	6.7	7.3	10.7
Forced Coordination	SP	0.7	14.7	2.0
Forced Coordination	PPO_{BC}	38.0	43.3	36.0
Forced Coordination	FACET	38.7	34.0	33.3
Counter Circuit	BC	16.0	17.3	19.3
Counter Circuit	SP	0.0	22.0	14.0
Counter Circuit	PPO_{BC}	7.3	2.7	6.0
Counter Circuit	FACET	3.3	22.0	12.7